A Comprehensive Review on Applications of Raspberry Pi in Engineering and IoT

Abstract

Raspberry Pi is an invaluable and popular prototyping tool in scientific research for experimenting with a wide variety of ideas, ranging from simple to complex projects. This review explores how Raspberry Pi is used in various studies, discussing its pros and cons along with its applications in home automation, healthcare, agriculture, and industrial robotics.

1. Introduction

The Raspberry Pi is a small but versatile prototyping device that has revolutionized scientific research and education. Its low cost, small size, and ease of use make it an ideal platform for a wide range of scientific applications, including data acquisition, control systems, and modeling.

Prototyping is an essential step in the design and development process, as it allows for the early identification of potential issues and promotes a fail-fast philosophy. Single board computers provide the computational backbone for modern cyber-physical implementations.

2. Proposed Methodology and Architecture

In our proposed system, multiple digital sensors interface with Raspberry Pi GPIO pins. We utilize MQTT protocols for real-time telemetry and a convolutional neural network for edge computer vision classification.

3. Implementation and Code Listing

The core sensor reader script is written in Python using OpenCV and PySerial libraries:

```
        import cv2
        import serial
        def capture_telemetry():
            cam = cv2.VideoCapture(0)
            ret, frame = cam.read()
            return frame
```

4. Conclusion

The Raspberry Pi offers a cost-effective, easy to use, and powerful tool that can be used for a variety of different tasks and projects.

References

[1] S. McManus, M. Cook, Raspberry Pi For Dummies, Wiley, 2021.
[2] H. K. Kondaveeti, Prototyping with Raspberry Pi in healthcare domain, IJACSA 2020.